

# Playing with the Rubik's Cube and its God's number in Rocq

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**Abstract.** One needs no more than 20 turns to solve any Rubik's cube position. This is called God's number and was computed in 2010. This note presents 3 smaller results about the Rubik's cube that are proved in Rocq. First, we prove that 20 is a lower bound. Second, we change metric (counting half turns as 2 moves) and prove that in this case 26 is a lower bound. Finally, the published computation of God's number (35 CPU years) splits the cube into the 2 217 093 120 cosets of a subgroup, one search to a coset. We formalise the correctness of the coset search, and certify the computation on one such coset.

**Keywords.** Rubik's cube, God's number, formal proof, Rocq, group theory.

## 1 The problem

A Rubik's cube is built from 26 small cubes: 8 corners with 3 stickers each, 12 edges with 2 stickers each and 6 centres. The centres do not move. A turn moves 4 corners and 4 edges, and rotates a centre.

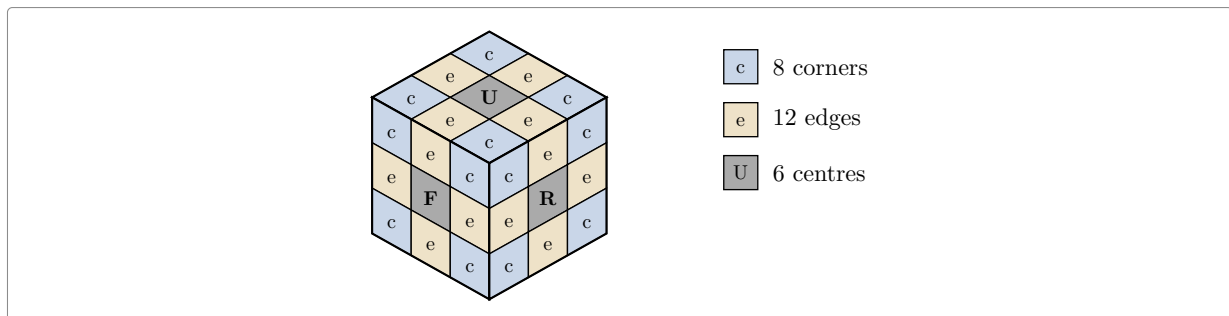


Figure 1: Anatomy of the cube

Not every arrangement of the pieces can be reached by turning faces. The number of possible arrangements is

$$8! \cdot 3^7 \cdot 12! \cdot 2^{11} / 2 = 43\,252\,003\,274\,489\,856\,000 \approx 4.3 \cdot 10^{19}.$$

The 8 corners can be in any order ( $8!$ ). Their orientations are free, except for the last one ( $3^7$ ). The same holds for the 12 edges ( $12!$  and  $2^{11}$ ). The result is divided by 2, because corners and edges are not independent.

Turning one face is a *move*. A half turn counts as one move, like a quarter turn. The number of moves needed by the worst position is called *God's number*. Counting a half turn as one move is a choice. A half turn can also count as 2 moves. This gives a second number for the same cube. We prove a lower bound for each.

We take one position: the **superflip**. Every small cube is at its right position, but all the edges have the wrong orientation.

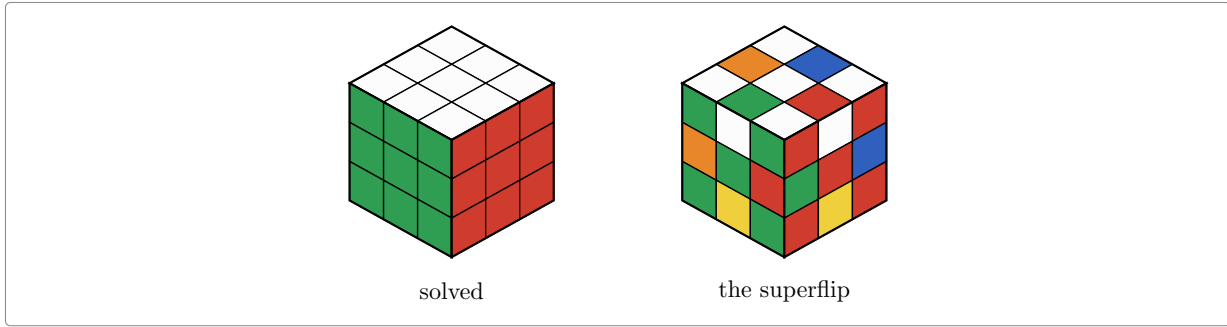


Figure 2: A solved cube, and the superflip.

The superflip is left unchanged by all 48 symmetries of the cube. We use it to prove the lower bound.

## 2 The cube as permutations

There are several ways to represent the Rubik's cube. We use stickers. There are 6 faces of 9 stickers. We remove the 6 centre stickers, which never move. The other 48 stickers are numbered from 0 to 47, as in Figure 3 and Figure 4. A move is then a rearrangement of the 48 stickers. Each face has 8 of them, taken left to right and top to bottom, with the centre skipped. Up gets 0–7, left 8–15, front 16–23, right 24–31, back 32–39 and down 40–47.

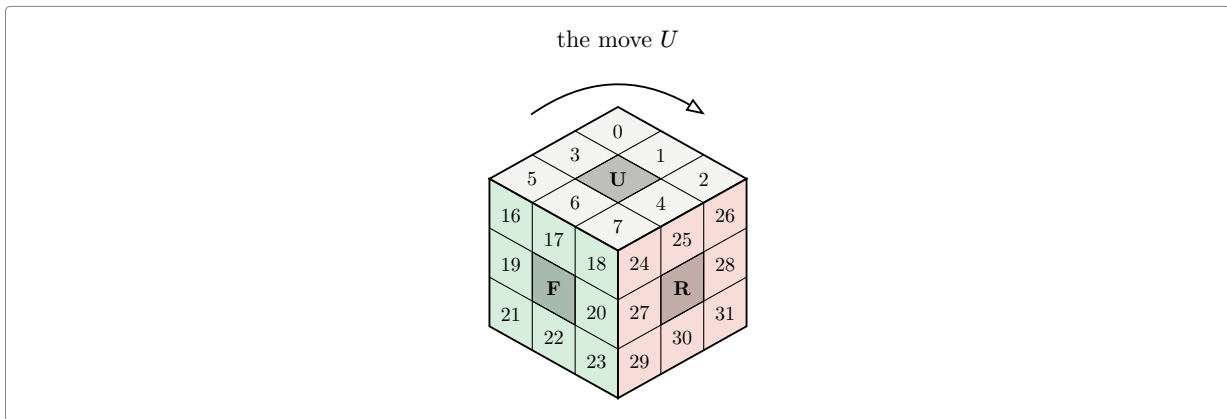


Figure 3: The top, front and right faces, and the turn of the top one.

|    |          |    |    |          |    |    |          |    |
|----|----------|----|----|----------|----|----|----------|----|
|    |          |    | 0  | 1        | 2  |    |          |    |
|    |          |    | 3  | <b>U</b> | 4  |    |          |    |
|    |          |    | 5  | 6        | 7  |    |          |    |
| 8  | 9        | 10 | 16 | 17       | 18 | 24 | 25       | 26 |
| 11 | <b>L</b> | 12 | 19 | <b>F</b> | 20 | 27 | <b>R</b> | 28 |
| 13 | 14       | 15 | 21 | 22       | 23 | 29 | 30       | 31 |
|    |          |    | 40 | 41       | 42 |    |          |    |
|    |          |    | 43 | <b>D</b> | 44 |    |          |    |
|    |          |    | 45 | 46       | 47 |    |          |    |

Figure 4: The cube unfolded, with all 48 places numbered.

A move sends each sticker to a new place. For example, if we turn the top face clockwise, the sticker in corner 0 goes to corner 2, the one in 2 to 7, the one in 7 to 5 and the one in 5 back to 0. This

is a cycle of 4 steps, written (0 2 7 5). The 4 edge stickers of that face do the same: (1 4 6 3). The turn also moves the top row of each side face to the next side face: front to left, left to back, back to right, right to front. This gives 3 more cycles, (8 32 24 16) and 2 similar ones. Figure 5 shows the whole move. Each square shows the sticker that is there after the turn. For example, the top row of the left face now holds 16, 17 and 18, the stickers that came from the front.

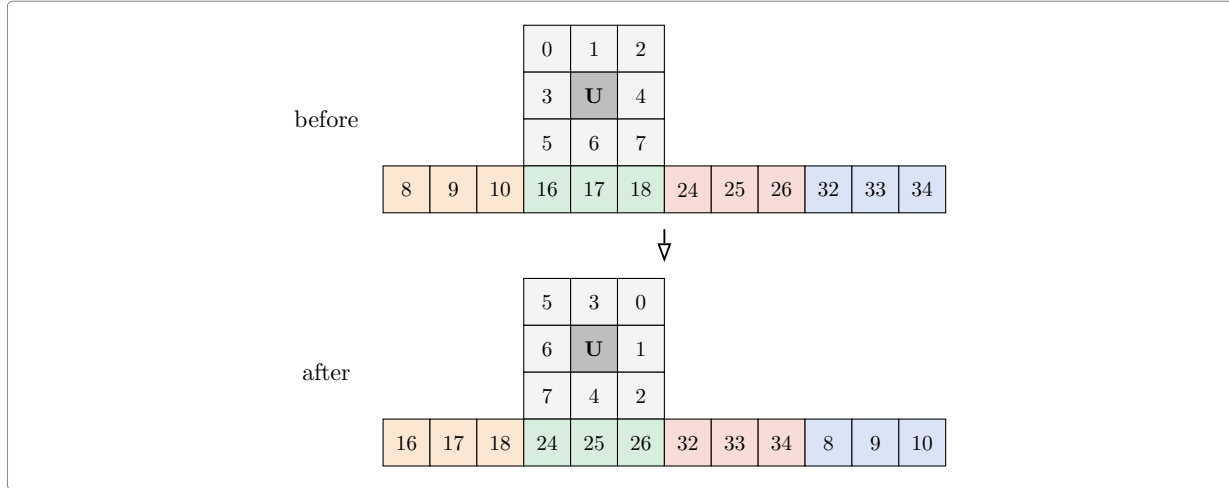


Figure 5: Everything a clockwise turn of the top face moves.

There are 6 clockwise quarter turns: up, right, front, down, left and back. Each of them can also be done twice or backwards. This gives the *18 moves*:

|    |    |    |    |    |    |
|----|----|----|----|----|----|
| U  | R  | F  | D  | L  | B  |
| U2 | R2 | F2 | D2 | L2 | B2 |
| U' | R' | F' | D' | L' | B' |

A move done *twice* is written U2, and a move done *backwards* is written U'. A position is a product of moves, written as a *word*. For example, R U R' U' is a word of length 4. The set of all positions is a group  $G$ , the *cube group*. A position solved in  $d$  moves is a word of  $d$  moves. The positions solved in at most  $d$  moves form the *ball of radius  $d$*  around the solved cube. God's number is the diameter of the Cayley graph.

## 2.1 The cube in Rocq

We use the Rocq prover [1] and its Mathcomp library [2]. The file `Rubik333.v` is a direct translation of the paragraphs above:

Definition facelet := 'I\_48.

Definition Umove : {perm facelet} := cyc [:: 0@; 2@; 7@; 5@] \* cyc [:: 1@; 4@; 6@; 3@] \* cyc [:: 8@; 32@; 24@; 16@] \* cyc [:: 9@; 33@; 25@; 17@] \* cyc [:: 10@; 34@; 26@; 18@]. (\* ... and five more, one per face ... \*)

Definition faces : seq {perm facelet} :=  
[:: Umove; Rmove; Fmove; Dmove; Lmove; Bmove].

Definition moves : seq {perm facelet} :=  
flatten [seq [:: g; g ^+ 2; g ^-1] | g <- faces].

Definition G : {group {perm facelet}} := <<Sset>>.

Line by line:

- 'I\_48 is the type of numbers below 48. It is a finite type.
- {perm facelet} is the type of *permutations* of those facelets.

- `cyc [:: 0@; 2@; 7@; 5@]` is the *cycle* that sends 0 to 2, 2 to 7, 7 to 5 and 5 back to 0, leaving the other 44 places where they are. The @ is local notation turning a plain number into a facelet.
- The symbol `*` composes 2 permutations. So `Umove` is the cycles of Figure 5 put together. `g ^+ 2` is the turn done twice, and `g ^-1` the turn undone. The order of composition is the opposite of the usual one. Mathcomp applies permutations on the right, so `(g * m) f` is `m (g f)`.
- `seq` is a list. `faces` is the list of the 6 clockwise quarter turns. `moves` takes 3 moves for each face, which gives the 18 moves.
- `<<Sset>>` is the group generated by the moves: all the positions that we can reach by composing moves. This is the cube group.

The superflip is defined in `Diameter.v`. It is made of 12 swaps, one for each edge, that exchange the 2 stickers of the edge:

```
Definition Spcyc : seq (seq facelet) :=
  [:: [:: 1@; 33@]; [:: 3@; 9@]; [:: 4@; 25@];
    (* ... nine more, one per edge ... *) ].
```

```
Definition superflip : {perm facelet} := \prod_(l <- Spcyc) cyc l.
```

This makes it a permutation, but not yet a legal position. For this, we have to prove that it belongs to  $G$ . The proof is one equality, with this word of 20 moves:

U R2 F B R B2 R U2 L B2 R U' D' R2 F R' L B2 U2 F2

## 3 Searching for the lower bound

We want to show that the superflip cannot be solved in 19 moves or less. The simple approach is to try every word of at most 19 moves from the superflip. If the solved cube never appears, 20 is a lower bound. But a tree of depth 19 with 18 branches at each node has  $18^{19}$  words. This is far too many. We need to refine the search.

### 3.1 The pruning estimate

A first refinement is a test that cuts early the branches that cannot succeed. For this, we give each position a rough lower bound  $h$  on the number of moves still needed. If  $h$  is 20 and only 18 moves remain, the branch cannot reach the solved cube in time. It is cut, with everything below it, as Figure 6 shows.

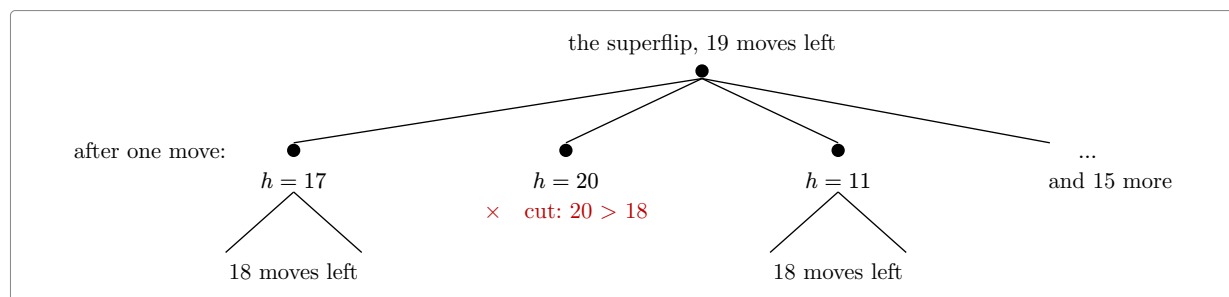


Figure 6: The search, and its scissors.

Such a lower bound is called an **admissible** estimate. A depth-first search that increases its depth step by step and cuts with such an estimate is Korf's IDA\* [3].

### 3.2 Getting a cheap estimate

The idea behind our estimate is to forget most of the cube. We keep only part of the information, for example how the corners are twisted and where the 4 middle-layer edges are. We call what is left a *summary*. Many positions share the same summary, and they get the same estimate. Moves act on summaries as well as on cubes. There are few summaries, so we can precompute the exact distance

of each one to the solved summary, and store it in a table. This gives  $h$ : take a position, compute its summary and read its distance in the table. The idea comes from Culberson and Schaeffer, who call such a table a *pattern database* [4]. Korf solved the cube optimally with 3 of them [5].

We use Kociemba's summary, from his two-phase solver [6]. We call it the *phase 1 summary*. Figure 7 shows the 3 things it records. A corner has one sticker of the up or down face. This sticker is in one of 3 places, numbered 0, 1 or 2. An edge is either the right way round or turned over, numbered 0 or 1. The 4 edges of the middle layer are in 4 of the 12 edge slots. The figure shades these slots on the 2 visible faces, where 3 of the 4 can be seen. Nothing else is recorded.

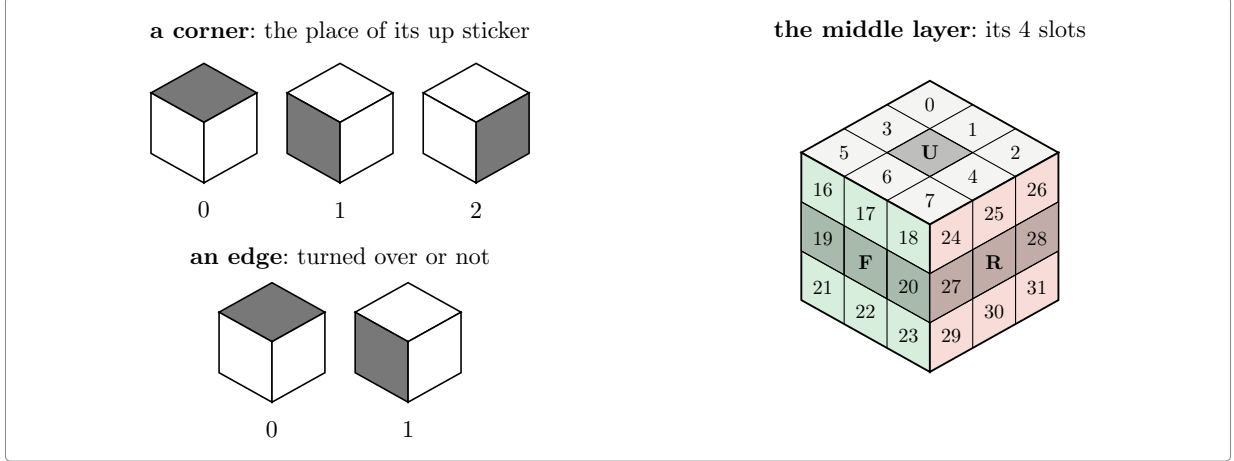


Figure 7: The 3 things a summary records.

The summary is the product of the 3 values:

| summary                            | values               |                   |
|------------------------------------|----------------------|-------------------|
| how the 8 corners are twisted      | 2 187                | $= 3^7$           |
| how the 12 edges are flipped       | 2 048                | $= 2^{11}$        |
| where the 4 middle-layer edges sit | 495                  | 4 places among 12 |
| <b>the summary</b>                 | <b>2 217 093 120</b> |                   |

Every summary stands for exactly 19 508 428 800 positions. The table gives, for each summary, its distance from the solved summary. A distance is never more than 12, so 4 bits hold one entry, and a 63-bit machine word holds 15 entries. The whole table is *1.18 GB*. The cut is very effective. A search at depth 14 visits only 445 398 nodes. Without the cut, the same tree has  $1.03 \cdot 10^{15}$  nodes.

### 3.3 Searching in Rocq

The search is generic. It is defined in the file `Search.v`. It takes a group, a set of moves, an estimate  $h$  and 2 hypotheses on  $h$ :

Hypothesis `h1` :  $h\ 1 = 0$ .

Hypothesis `hstep` :  $\text{forall } g\ m, m \in S \rightarrow h\ g \leq (h\ (g * m)).+1$ .

Fixpoint search ( $d : \text{nat}$ ) ( $g : gT$ ) :  $\text{bool} := (h\ g \leq d) \ \&\& \ ((g == 1) \ || \ (\text{if } d \text{ is } d'.+1 \text{ then has } (\text{fun } m \Rightarrow \text{search } d' (g * m)) \text{ Sseq else false}))$ .

Corollary `searchN d g` :  $\text{search } d\ g = \text{false} \rightarrow g \notin \text{ball } S\ d$ .

Line by line:

- `gT` is the group. It is then instantiated with the cube group.
- `1` is the unit of that group, the solved position for the cube. So with  $h\ 1 = 0$  we ask the estimate to be 0 on the solved cube, and  $g == 1$  tests whether the search has reached the solved position.

- `Sseq` is the list of the 18 moves, and `S` is the same thing seen as a set.
- `g * m` is the position `g` followed by the move `m`.
- `h g <= d` is the cut. `(h (g * m)).+1` is the estimate after `m` plus one, so by `hstep` one move changes the estimate by at most one.
- `has (fun m => search d' (g * m)) Sseq` tries every move with one fewer move left, and answers as soon as one of them succeeds.

The last line is what we need. If the search returns false, the position is not in the ball of radius  $d$ . The next subsections describe the improvements that give our final search.

### 3.4 Searching with a summary

The estimate is built in a second generic file, `Coord.v`. It takes the summary of a position, the action of a move on a summary and the table of distances. It asks one condition on the summary and 2 on the table.

```
Variable coord : {perm facelet} -> X.
Variable act    : X -> {perm facelet} -> X.
Hypothesis coordM : forall g m, coord (g * m) = act (coord g) m.
```

```
Variable D : X -> nat.
Hypothesis D0 : D (coord 1) = 0.
Hypothesis Dstep : forall x m, m \in Sset -> D x <= (D (act x m)).+1.
```

`X` is the type of summaries, and `coord` gives the summary of a position. `act` applies a move directly to a summary, without going back to the position. By `coordM`, the two agree: applying a move and then taking the summary gives the same result as `act`.

The search of `Search.v` is refined to carry the summary with the position. The summary is updated with `act` at each move. The position is still needed, to test whether the cube is solved. Here is the refined search, with simpler names and some details left out. The real one is `searchz3` in `Farpl.v`.

```
Fixpoint search (d : nat) (g : gT) (x : summary) (p : move) : bool :=
  (D x <= d) &&
  ((g == 1) ||
   (if d is d'.+1
    then has (fun m => search d' (g * m) (act x m) m) (allowed p)
    else false)).
```

Here are some explanations:

- `D x <= d` is the cut. It reads the table at the summary `x`.
- `g == 1` tests whether the cube is solved.
- `g * m` moves the position and `act x m` moves the summary. The summary never has to be recomputed from the position.
- `p` is the last move, and `allowed p` is the list of moves allowed after it. This is explained in section 3.6.

### 3.5 Verifying the table

For the phase 1 summary, `D` is a lookup in the phase 1 table, so `D0` and `Dstep` become 2 statements about that table:

- the entry of the solved summary is 0;
- every entry is at most one more than the entry reached from it by any of the 18 moves.

The table is produced by an OCaml program, and it gives the exact distance. But the table is not trusted. The 2 statements above are checked in Rocq by computation. A table full of zeros would also pass these checks, but then the search would cut nothing. The entries of the table do not depend on each other. So the second check, the most expensive one, is cut into slices, one file for each slice. The files are checked in parallel, since Rocq compiles files separately. This takes 10 minutes. All the

timings in this note are measured on the same machine, the *reference machine*: a dual-socket Intel Xeon E5-2667 at 2.9 GHz, with 12 cores, 24 threads and 62 GB of memory.

### 3.6 Removing redundant moves

Many words lead to the same position. We use 2 ideas to leave some of them out. The first idea is to avoid repetition. For example, after a  $U$  we do not need to try  $U$ ,  $U2$  or  $U'$ . Shorter words, at a smaller depth, already reach these positions. This leaves 15 moves instead of 18. Opposite faces give a weaker form of the same argument.  $U D$  and  $D U$  give the same position, so we keep only one of the 2 orders. We privilege the top, right or front face first. We call this the *order convention*. From the second move on, it leaves 15 moves after a turn of the top, right or front face, and 12 after a turn of the bottom, left or back face. The second idea is symmetry. The superflip is unchanged by all 48 symmetries of the cube. So for the first move we only need the turns of one face. We choose the top face. By symmetry again, we only need  $U$  and  $U2$ , since  $U'$  is the mirror image of  $U$ . The search is then parallelised at depth 2: 2 first moves times 15 second moves is 30 *prefixes*, and each prefix is searched in a file of its own.

### 3.7 Composing summaries

We can turn the whole cube around the axis through 2 opposite corners. We get the same position, seen from another side. Its summary gives another entry of the same table. So each position has 3 summaries, and the maximum of the 3 values is still an admissible estimate. This costs 3 reads of the table instead of one. But it is worth it, because the tree is much smaller.

### 3.8 Folding the table

The summary is built around the up-down axis. Of the 48 symmetries, 16 keep this axis. They turn one summary into another. They sort the 1 013 760 values of flip and slice into 64 430 families. One entry for each family is enough, so the table can be made smaller. In the code, the change is one definition. The lookup was

```
Definition Dpli (tw x : int) : int := plget (plidx tw x).
```

and it became

```
Definition Dplri (tw r : int) : int :=
  plget (plfoldi (frep r) (twsym tw (fsym r))).
```

`frep r` is the representative of the family. `fsym r` is the symmetry that sends the value to it. `twsym tw (fsym r)` is the twist sent by the same symmetry. Each lookup costs 3 more reads, in a table 15.73 times smaller.

Tables reduced by symmetry are common in cube solvers. With the fold, the search is 1.61 times slower at depth 16. But a search worker goes down from 4.15 GB to 3.6 GB, so 18 of them can run at the same time. And checking the table goes down from about 5.4 processor hours to 1.35.

### 3.9 Conclusion on the first lower bound

The development for this lower bound has 44 files: 9 about the cube (2 500 lines), 15 about the search (4 900 lines) and 20 about the tables (5 200 lines). Building the tables costs the same for any radius of the search. Here are the times, measured from a clean tree on the reference machine:

|                                                       | wall clock | processor time |
|-------------------------------------------------------|------------|----------------|
| emitting the tables and compiling them to native code | 17 min 21  | 52 min 13      |
| the coordinate and summary tables                     | 22 min 46  | 21 min 15      |
| the search and the estimate, over no data at all      | 11 min 02  | 30 min 45      |
| the 4 certificates for the move and distance tables   | 56 s       | 2 min 27       |
| the fold: 12 checks and 27 slices                     | 9 min 52   | 47 min 57      |
| the shape check against the dummy table               | 18 s       | 16 s           |

|                 | wall clock    | processor time |
|-----------------|---------------|----------------|
| <b>in total</b> | <b>1 h 02</b> | <b>2 h 35</b>  |

The second line runs on one core, so more cores do not help. The first and fifth lines are mostly the OCaml compiler, turning a table into native code. Together they take 100 of the 155 processor-minutes.

The search visits 137 607 893 106 positions. The smallest piece has 3 067 879 204 and the largest 8 527 685 275. Between depths 17 and 19, the tree grows by 12.22 from one level to the next. We measured the run twice, once before the fold and once after. The theorem is the same both times.

| radius 19, search depth 17 | before  | after          |
|----------------------------|---------|----------------|
| pieces                     | 18      | <b>30</b>      |
| workers                    | 9       | <b>18</b>      |
| memory per worker          | 4.15 GB | <b>3.6 GB</b>  |
| wall clock                 | 11 h 13 | <b>5 h 48</b>  |
| processor time             | 85 h 11 | <b>89 h 27</b> |

The 30 pieces balance the run. Each piece is one prefix of 2 moves. They are compiled longest first, since make starts them in the order it is given. The pieces are far from equal. The longest takes 5 h 43 of processor time, and the shortest 1 h 43. So the longest piece sets the wall clock. The wall clock, 5 h 48, is only 5 minutes more than the best that any order of these 30 pieces can give.

## 4 Counting in quarter turns

We have proved that God's number is at least 20. We now count the moves differently. In quarter turns there are 12 moves: the 6 faces turned one way, and the same 6 turned back. A half turn counts as 2 moves. In this count, the answer is **26** (cube20.org).

### 4.1 The position

The superflip is only 24 quarter turns from solved, so it is not far enough. Reid posted a better position to the Cube-Lovers list in August 1998 [7]: the superflip composed with four spot, the four-spot pattern with the superflip on top of it. In the following, we call it **superflip4**.

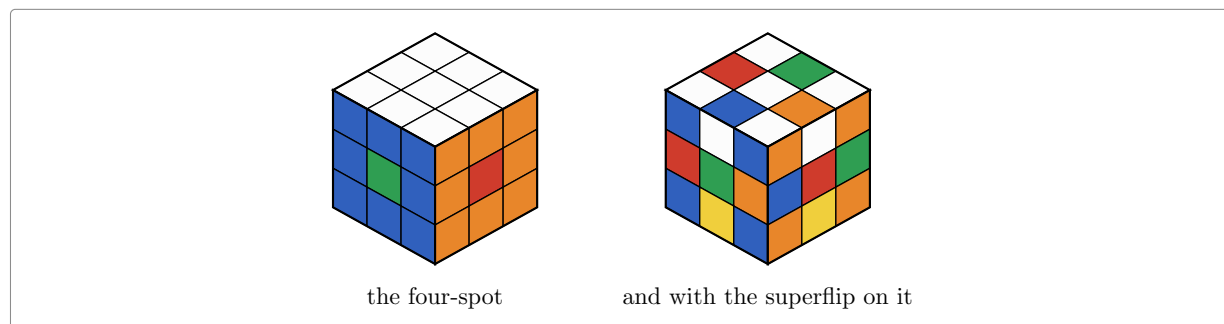


Figure 8: The four-spot, and superflip4.

Superflip4 is 26 quarter turns from solved. Here is the word given for it, with the half turns written out:

**U U D D L F F U' D R R B U' D' R L F F R U D' R' L U F' B'**

We want to prove that this position cannot be solved in 25 quarter turns. The parity of permutations saves one level of search (24 instead of 25). A cycle of 4 stickers is the product of 3 swaps, so it is an odd permutation. A quarter turn is 5 cycles of 4 stickers, so it is odd too. So a word of even length gives an even permutation, and a word of odd length gives an odd one. This argument is in **HBound.v**.

Because of this word of 26 moves, superflip4 is even. So all its words have an even length. If superflip4 has no word of at most 24 moves, it is exactly 26 moves from solved.

## 4.2 The 6 prefixes

Every shortest word for superflip4 can be rewritten, at the same length, so that it begins with one of 6 prefixes [7]:

$$\begin{array}{ccc} \text{R U} & \text{R' U D} & \text{R' U F'} \\ \text{R' U R'} & \text{R' U B'} & \text{R' U L'} \end{array}$$

The searches start from those 6. In order to prove this fact, we split the 12 quarter turns into 2 sets:

$$\mathcal{A} = \{ \text{U, U', D, D'} \}, \quad \mathcal{C} = \{ \text{R, R', F, F', L, L', B, B'} \}.$$

A word made only of turns from  $\mathcal{A}$  leaves the 4 middle edges untouched, and superflip4 has them flipped, so no such word gives superflip4. A word made only of turns from  $\mathcal{C}$  never flips an edge, and superflip4 has every edge flipped, so no such word gives it either. Any word for superflip4 therefore uses turns from both sets, so somewhere in it there is a two-letter subword made of a letter of  $\mathcal{A}$  and a letter of  $\mathcal{C}$ .

Each of the 6 prefixes starts with a turn from  $\mathcal{C}$  followed by a turn from  $\mathcal{A}$ , so we want a word for superflip4 that starts with such a subword. Recall the word we gave for it:

$$\text{U U D D L F F U' D R R B U' D' R L F F R U D' R' L U F' B'}$$

Reading along it, 4 two-letter subwords have the right order and 5 have the wrong order. We take the first one with the right order,  $\text{F U'}$ , and call it  $q$ . A word may have had only subwords with the wrong order. In that case, we would then invert it first: superflip4 is its own inverse, so the inverse is again a word for it, of the same length, and the orders are swapped. Here the inverse is

$$\text{B F U' L' R D U' R' F' F' L' R' D U B' R' R' D' U F' F' L' D' D' U' U'}$$

with, as expected, 5 subwords in the right order and 4 in the wrong order.

Let us go back to our two-letter subword  $q$ . It has 6 turns before it. Write  $x$  for them,  $\text{U U D D L F}$ , and  $w$  for what follows  $q$ , so that  $xqw = P$ , where  $P$  is superflip4. Then  $qw = x^{-1}P$ , and therefore

$$qw(P^{-1}xP) = x^{-1}PP^{-1}xP = P.$$

$P^{-1}xP$  is not longer than  $x$ :

$$P^{-1}xP = (P^{-1}\text{U}P)(P^{-1}\text{U}P)(P^{-1}\text{D}P)(P^{-1}\text{D}P)(P^{-1}\text{L}P)(P^{-1}\text{F}P) = \text{U U D D R B}.$$

The first step is done:  $qw$  followed by those 6 turns is a word for superflip4 of the same length, beginning with  $q$ .

$$\text{F U' D R R B U' D' R L F F R U D' R' L U F' B' U U D D R B}$$

The second step uses symmetries. A symmetry of the cube that leaves superflip4 unchanged carries a word for it to another word for it, of the same length. Of the 48 symmetries, 16 do that. The superflip is unchanged by all 48, but the four-spot is not: it leaves the top and bottom faces alone and exchanges the colours of the other 4 in pairs, so it singles out the up-down axis, the line through the centres of the top and bottom faces. A symmetry that moves that axis carries the four-spot to the same pattern about another axis, which is another position. The 16 that keep the axis are the ones that leave superflip4 alone. They make the letter of  $q$  from  $\mathcal{A}$  into  $\text{U}$ , and the letter from  $\mathcal{C}$  into  $\text{R}$  or  $\text{R'}$ , so  $q$  becomes  $\text{R U}$  or  $\text{R' U}$ . In our example  $q$  is  $\text{F U'}$ . The mirror that swaps left and right fixes the front face and reverses the direction of every turn, so  $\text{F U'}$  becomes  $\text{F' U}$ . A quarter rotation about the up-down axis then carries the front face to the right one, and  $\text{F' U}$  becomes  $\text{R' U}$ .

A word starting with **R U** needs nothing more, which is the prefix of 2 turns. When it starts with **R' U** we look at the third turn. For 6 of the possible turns, we get words that a symmetry or an inversion brings back to the **R U** case, and 5 do not. Those 5 are the prefixes of 3 turns.

### 4.3 The summary, and its table

The estimate is built as before. We take the summary Reid uses in the quarter-turn count. Against the phase 1 summary of the first bound, it keeps the corner twist, the same 2 187 values, and replaces the edge flips and the slice by where the 4 middle edges sit with their flips, and by which 4 corner places hold the top corners.

| summary                                                                     | values                |                                   |
|-----------------------------------------------------------------------------|-----------------------|-----------------------------------|
| where the 4 middle-layer edges sit, each of them the right way round or not | 190 080               | = $24 \cdot 22 \cdot 20 \cdot 18$ |
| which 4 corner places hold the 4 top corners                                | 70                    | 4 places among 8                  |
| how the 8 corners are twisted                                               | 2 187                 | = $3^7$                           |
| <b>the 3 together</b>                                                       | <b>29 099 347 200</b> |                                   |

Each of the 4 edges can sit in any free slot, either way round. That gives 24 choices for the first, and 2 fewer for each of the others, since a slot taken is taken whichever way round the edge in it lies. A summary is the coset of a subgroup H of the cube group, which is why the file names of this section start with H. It is not the subgroup of the first bound, and the summaries are 13 times as many, 29 billion against 2.2 billion.

The table holds the distance from solved of each of the 29 billion summaries. The number of summaries at each distance agrees with the column published in 1998, and we check that first. The table is then folded. The 16 symmetries that keep the up-down axis sort the 190 080 edge values into 12 094 families, a factor of 15.72, and one entry is kept per family. That is 883 MB, and 3.86 GB once loaded into the prover.

### 4.4 Conclusion on the second lower bound

The word of 26 moves and the 6 searches together show that superflip4 is exactly 26 quarter turns from solved. So in quarter turns, God's number is at least 26. The development for this lower bound has 19 files (6 100 lines). The argument for the 6 prefixes is in **HProp2.v**, the search in **HSearch.v**, the checks in **HSweep.v** and the bound in **HAL1.v**. Here are the times, measured from a clean tree on the reference machine:

|                                       | wall clock     | processor time |
|---------------------------------------|----------------|----------------|
| building the table, in OCaml          | 9 min 50       | 1 h 43         |
| the same table as 59 Rocq files       | 3 h 13         | 6 h 50         |
| checking the 3 move tables            |                | 1 min 20       |
| checking the distance table, 12 jobs  | 2 h 51         | 31 h 42        |
| the 6 searches, 72 pieces, 12 workers | 4 h 00         | 45 h 54        |
| <b>the whole chain in Rocq</b>        | <b>10 h 32</b> | <b>87 h 29</b> |

The last row is measured end to end, from a directory where nothing is built. It is not the sum of the others. The OCaml table of the first row is built once by hand, and it is not part of that run. The checks and the searches are nearly nine tenths of the cost. Checking the table, instead of trusting it, costs about two thirds of what the searches cost.

## 5 One coset of the upper half

So far we have dealt with lower bounds. This section is about the upper bound. Every position of the cube is solved in 20 moves or fewer.

Take a subgroup  $H$  of  $G$ . For a position  $x$ , the *coset*  $xH$  is the set of the positions  $xh$  with  $h$  in  $H$ . Every position lies in exactly one coset. We take one position  $x$  in each coset. It is the *representative* of the coset. It is then enough to show that every position of  $xH$  is solved in 20 moves or fewer.

For one coset, we use a *marking*. Each position of  $xH$  gets one bit. This bit is initially set to 0. The marking works iteratively. For each level  $d$ , from 0 to 20, we list the words of length  $d$ . A word gives a position. If this position is in  $xH$ , we set its bit to 1. At the end, the bits set to 1 are exactly the positions of  $xH$  within 20 moves. If no bit is left at 0, every position of the coset is solved in 20 moves or fewer. Doing this for every representative gives the upper bound.

This is the main idea: the marking is local to one coset. The work on one coset does not depend on the work on another. So the problem splits into many independent problems. They can be done in parallel. A coset has as many positions as  $H$ , with one bit for each. So if  $H$  is not too large, the memory needed is small.

The choice of  $H$  is crucial. The next subsection presents the one we take.

## 5.1 Cosets

We have already seen the subgroup  $H$  we take. It is the set of the positions whose phase 1 summary is the solved one. A coset of  $H$  is one value of the summary. There are as many cosets as summaries.

$H$  has 19 508 428 800 positions. So a coset is 19 508 428 800 bits, about 2.4 GB. This fits in the memory of a desktop machine. There are 2 217 093 120 cosets. So up to 2 217 093 120 problems can be run in parallel.

|                       | count                      |
|-----------------------|----------------------------|
| positions of the cube | 43 252 003 274 489 856 000 |
| cosets                | 2 217 093 120              |
| positions in a coset  | 19 508 428 800             |

We use 2 facts from section 3.

- **Checking membership is quick.** A position  $p$  is in  $xH$  exactly when  $x^{-1}p$  is in  $H$ , that is, when the summary of  $x^{-1}p$  is the solved one. The bit of a position  $xh$  of the coset is indexed by  $h$ . So we start our enumeration of the words of length  $d$  from  $x^{-1}$ . We apply the  $d$  moves of a word. If the position  $h$  we reach is in  $H$ , we set the bit of  $h$ .
- **The distance to the solved summary is known.** The phase 1 table gives it. For a position, this is the number of moves needed to bring it into  $H$ . We use it to cut the enumeration, as in the search of section 3. We build the words of length  $d$  one move at a time. Say  $k$  moves have been applied, and the table gives  $t$ . If  $k + t > d$ , no word that continues from there can end in  $H$  after  $d$  moves. We drop this branch. The table is checked to be never too large (section 3.5). So a cut never drops a word that ends in  $H$ .

What is new is that  $H$  is generated by 10 of the 18 moves:  $U, U2, U', D, D2, D', R2, L2, F2$  and  $B2$ . This is a result of Kociemba. These are the moves that do not change the summary. So  $H$  splits the 18 moves in 2 sets. The 10 moves of  $H$  do not change the summary, so they do not change the distance to  $H$ . The other 8 moves,  $R, R', L, L', F, F', B$  and  $B'$ , change the summary. They can change the distance.

The published proofs do not solve all 43 quintillion positions one at a time. They solve a whole coset at once. One search settles every one of the 19 508 428 800 positions in it. The positions in a coset are not all the same distance from solved. None of them is more than 20 moves from solved. A symmetry of the cube carries one coset to another, and the image is solved by the same words transformed by the symmetry, so only one coset per symmetry class is searched. We then need 2 things. The cosets searched must cover every class, and each search must settle its whole coset.

`Canon.v` proves the first, and it is not a computation. As the representative of a class we take its least member, in the order the type of positions already carries. Every class is covered, because a finite set that is not empty has a least member. The file is 80 lines and assumes nothing.

The second is the computation itself. Rokicki, Kociemba, Davidson and Dethridge ran it over 55 882 296 families of cosets. It took about a billion seconds of CPU time, given by Google, which they count as 35 CPU years of a four-core desktop processor. We cannot repeat that. What we can do is one coset, to see what one costs and whether the pieces are in place. We did the superflip's, because we already had the superflip from the lower bound.

## 5.2 The coset as one map

A position of the coset is named by 3 numbers: how the 8 corners of the top and bottom layers sit, how the 8 edges of those layers sit and how the 4 middle edges sit. That is 40 320 by 40 320 by 24 arrangements, but half of those triples cannot occur: on the cube the corners and the edges are always permuted with the same sign. The map holds one bit for each of the 19 508 428 800 triples that are left.

The bits are laid out the way Rokicki's own program lays them out, and the layout is what makes the next subsection cheap. A page is one arrangement of the corners. Inside a page, a group is a pair of arrangements of the outer edges, the 2 that differ by exchanging 2 edges, and the group's 24 bits are the 24 arrangements of the middle edges, the 12 even ones low and the 12 odd ones high. The parity determines which arrangement of the pair a bit stands for, which is how the impossible half disappears with nothing left to store.

A machine word holds 48 bits, so one word holds the same group on 2 pages, the corner arrangements of even and odd rank, the odd one in the top half. The map is then 20 160 times 20 160 such words: 406 425 600 words, 3.25 GB. An array in Rocq may hold 4 194 303 entries, so the map is an array of 194 chunks of 2 million words. The map and its indexing are `Row.v` and `RowMap.v`.

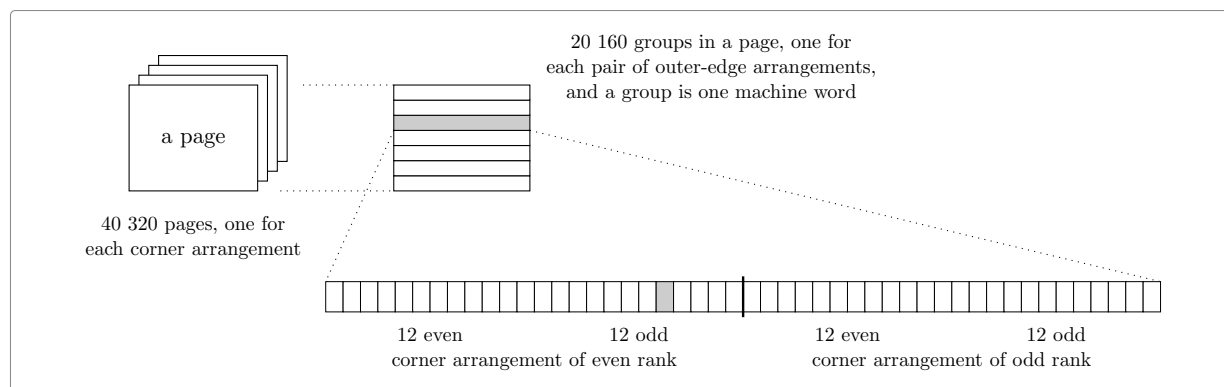


Figure 9: The map, from the outside in. A page for each arrangement of the corners, a group in the page for each pair of outer-edge arrangements and a bit in the group for each arrangement of the 4 middle edges, the 12 even ones low and the 12 odd ones high. One word holds the same group on the 2 pages of a pair.

The invariant the whole proof turns on is one line: a map is **sound at  $d$**  when every bit it has set is a member of the coset that is within  $d$  moves of solved. The map the run starts from has one bit set, the superflip's own, and is sound at nought. Each of the 20 steps below takes a map sound at  $d$  to a map sound at  $d + 1$ . If the map that comes out has every bit set, then every member is within 20 moves, and that is the theorem.

## 5.3 A level

The run is 20 levels, level  $d$  turning a map sound at  $d - 1$  into a map sound at  $d$ . A level does 2 things, and they divide the words of length  $d$  between them.

- The **prepass** applies each of the 10 moves of  $H$  to the whole map at once. It accounts for every word whose last move is in  $H$ , and that is nearly all of them.
- The **search** then looks only for the words of length  $d$  whose last move is not in  $H$ . It walks them from the superflip one position at a time, the way the searches of the earlier sections do.

In `RowSrch.v` the run is 7 lines. It carries the map, the map it reads while it writes and the number of bits the last level left, which is what decides whether the cuts below are on.

```
Fixpoint runsk (n : nat) (d : nat) (n0 : int) (m dst : rmap) : rmap :=
  if n is n1.+1 then
    let m' := levelsk (Uint63.ltb ncutb n0) d.+1 m dst in
    runsk n1 d.+1 (mcount m') m' m
  else m.
```

`levelsk` is the level: the prepass first, then the search if this level is still being searched. Counting the bits, `mcount`, is a traversal of the whole map, and the run pays for it once a level.

## 5.4 The prepass

The prepass is what makes a whole coset affordable, and it is the one part of the computation that has no counterpart in the lower bounds.

Applying a move of  $H$  to a position of the coset does 3 separate things to the 3 numbers that name it. The corner arrangement goes to another corner arrangement, so a page goes to a page. The outer-edge pair goes to another pair, so a group goes to a group. The middle arrangement goes to another middle arrangement, so the 24 bits of the group are rearranged among themselves, and that rearrangement depends only on the move and the group it came from. It is a table lookup and a shuffle of one machine word.

So the prepass never takes a position apart. It never builds a cube, never ranks one, never looks a position up. For each of the 10 moves it reads the whole map and writes the whole map, and that is all of it. The 10 passes over 3.25 GB cover every word of this length that ends in a move of  $H$ , and there are billions of those.

`RowMap.v` has the prepass and the proof that it keeps the map sound. `RowLvl.v` has it again, written so that a page's chunk is fetched once and put back once instead of once a word, and proves the two are the same function, so nothing about the cube is proved twice.

## 5.5 The search

The search is a depth-first walk from the superflip, like the ones in sections 3 and 4, with 2 differences. It carries the position it has reached, and when it has used up its depth it asks whether that position is a member of the coset. If it is, the position's 3 numbers are computed, and the bit they name is set. That is the leaf, and it is 2 lines of `RowSrch.v`:

```
else if csolved c
  then let: (pg, gr, bt) := plc (tomemb x) in mmark m pg gr bt
  else m
```

The membership test is made on the position itself, which the search carries anyway, and it costs one comparison. It is worth explaining why it is not made on the phase 1 table, which is right there and would give the same answer for a handful of nanoseconds less. The table is an estimate. A table of zeros is a legal estimate – it passes both conditions of section 3.5, so nothing rules it out – and with a table of zeros every position would look like a member, every bit would be set at once and the theorem would be empty. Reading membership off the table would make the theorem depend on the table being sharp, which nothing proves. Reading it off the position makes it depend on nothing.

The other use of the table is a cut, and that one is sound. At each node the search looks the position up and gets a number that is never larger than the number of moves needed to bring it into  $H$ . If

that number is larger than the moves left, the branch cannot reach the coset and is dropped. An estimate that is too small only cuts less than it could.

## 5.6 The cuts and the early stop

A search that offered all 18 moves at every node would never finish. It is cut down by 4 things, 3 of them Rokicki's.

The first is the moves worth trying. The phase 1 table of section 3.8 is already folded by the 16 symmetries; the copy the search reads, `RowMask.v`, carries beside each distance the set of moves that bring the position nearer  $H$  and the set that at least do not take it further. Which set is wanted depends on how many moves the search has to spare. A node then offers 3 or 4 moves instead of 18. The entry is 28 bits, 2 to a machine word: 4 bits of distance and 4 for each of the 6 faces. Written out one bit a move it would take 41. A half turn is a quarter turn twice, so the 2 moves of one face cannot differ by more than one step, which leaves 15 of the 27 ways a face can go, and 15 fit in 4 bits.

The second is that the last move of a word is never a move of  $H$ . A word that ends in  $H$  is a shorter one followed by moves of  $H$ , and the prepass has already applied those.

The third is that low down, a move must go straight at  $H$ . Once the moves left and the distance to  $H$  add up to less than 5, only the moves that shorten the distance by one are followed. A move that wastes a step down there ends in moves of  $H$ , so the prepass catches that word too.

The fourth is where the search stops altogether. It runs on the first 16 levels and not on the 4 above them, and on level 16 it does not even run out: it stops as soon as the map holds 167 million bits plus a third of what that level's prepass left. Those are Rokicki's numbers, and every published time of his was measured with them on. What is left is finished by the prepasses of levels 17 to 20.

The second and third cuts only come on once the map holds more than 6 million bits, which on this coset is not before level 14, and that too is his rule.

Every one of these 4 is safe, and safe in a way none of the cuts in the earlier sections were. There, a cut that lost a word would have lost the proof, and each one had to be shown to lose nothing. Here nothing is ever proved about what the search covers. What is proved is that the map filled. A cut that throws a word away can only make the map fill later, or leave a bit clear at the end – it can never set a bit that should not be set. This is why a coset is easier to prove than a lower bound, and it is why the 4 cuts above needed no argument at all. The one thing that did need an argument is marking, which is the subsection above.

## 5.7 The members left over

Stopping the search at level 16, and stopping level 16 early, leaves members the run never reaches. When the 2 runs ended, 32 bits of the 406 425 600 words were still clear. They are simply the members that no word the cut search kept, followed by moves of  $H$ , happened to reach. Searching deeper would leave fewer and cost hours; each one left over costs a line.

So we give each of the 32 a word of 20 moves by hand, in `RowWits.v`. Our prototype gave 28 of them, and the last 4 were found one at a time. None of that is believed. `RowWitsChk.v` applies each of the 32 words to its member and asks for the solved cube. If the producer had made a mistake, or given a word of 21 moves, or named the wrong member, the check would fail; it cannot make the theorem false. The replay is also the only thing in the development that tests the 2 ideas of which facelet is which – the prototype's, which found the words, and Rocq's, which builds the member – against each other.

The 32 are marked into the map at the very end, by `RowMark.v`, and not at the start. In a map sound at  $d$ , every bit set is within  $d$ . A witness is within 20, so seeding one before the run would make level one mark its neighbours as within one, and they are not. Marking at the end is sound because the map is already sound at 20 when the marks go in.

## 5.8 The run, and its one boolean

Everything above is one boolean. `RowFoldCubDefI.v` builds the map, runs the 20 levels, marks the 32 and asks whether every bit is set:

Definition `rowfulli : bool := mfullf ffuli ycwitsoi`.

Lemma `rowfulliE : rowfulli = true`.

Proof. `Time native_cast_no_check (erefl true)`. Qed.

That Qed is the 6 hours. The file that holds it, `RowFoldCubBoolI.v`, is one **Require** and one **Lemma**: a file that runs loads no proof, and a file that proves runs nothing. Everything else – that the prepass keeps the map sound, that the search marks only members, that a full map is the theorem – is proved beside it, of a map and a run the proof never evaluates.

## 5.9 Folding the map

The map can be made smaller, and we made it smaller to see what its size does to the run. Both maps were searched, the folded one and the unfolded one, and the 2 runs are compared at the end of the section.

The map is 40 320 pages, one for each arrangement of the corners.

The fold works because of the position we chose. Of the 48 symmetries of the cube, 16 keep the top and bottom faces in place. Each of them sends the 10 moves to the 10 moves, so it maps  $H$  to itself, and each leaves the superflip unchanged, so it maps the superflip's coset to itself. A member and its image under such a symmetry need the same number of moves, so 2 pages related by a symmetry hold the same answer. One page of each family is then enough: 2 768 of the 40 320, a factor of 14.6. A level of the run is one pass over the map, so there is 14.6 times less of it to walk. The price is undoing a symmetry whenever a kept page is read.

The kept pages go in pairs of their own, and a pair shares one word as it does on the unfolded side: 1 496 words in place of 2 768. There are 224 pages that are their own partner and use half a word.

The fold has to be proved as well as written. A symmetry sends a member of the coset to a member of the coset. Undoing it gives back the position the page stood for. A map sound after one level is sound after the next. That is the largest single part of the coset's proof.

## 5.10 The reused and the new

We reuse nearly everything. The abstract search and its contract come over unchanged, and so do the cube, the permutations, the machine-integer tools, the phase one table and its generator. Each of the new tables has a check of its own, in a file of its own, so that a failure names the table that failed.

There are 5 new things, and only one of them is a search.

- The rank and the sign of a permutation. Rocq's library has both, but for permutations that do not compute, so we need their effective version on machine integers.
- The link between a position and its 3 ranks. The map's bits are triples of numbers, so what the run proves is a statement about triples. To read it as a statement about the cube we have to turn a triple back into the position it stands for. The other way we already had.
- The pass that steps a whole map one move at a time.
- The fold by the 16 symmetries.
- Words given by hand for the members the run does not reach.

## 5.11 The files and the cost

The coset adds 70 hand-written files and 17 600 lines, besides the generated tables. The coset and its members are `Row.v`, `RowMemb.v` and `RowInst.v` the ranking and the moves `Lehmer.v` with the `RowPart`, `RowMove` and `RowTab` groups; the map and the search `RowMap.v`, `RowRun.v`, `RowSrch.v` and

RowFinal.v the fold the RowFold group; and the 2 runs the RowCub and RowFoldCub groups, ending in RowCubDoneI.v and RowFoldCubDoneI.v. README.md lists every one of them with what it does.

The search ran twice, over the folded map and over the unfolded one, with the same search in both, and both times it filled the map.

| the run               | wall clock | processor time | peak memory |
|-----------------------|------------|----------------|-------------|
| over the folded map   | 5 h 53     | 5 h 51         | –           |
| over the unfolded map | 7 h 52     | 7 h 48         | 23.3 GB     |

The fold is worth 1.3 times on the wall clock and 1.3 times on processor time. The run does not follow the size of the map, which is 13 times smaller. Most of the work is the search at the deepest levels, and that is the same tree on both sides. The map sets the memory. The unfolded map is 3.25 GB against 248 MB, and a level reads one map while it writes the other, so the unfolded run needed 23.3 GB of a 62 GB machine. With the garbage collector set to work harder, the same run peaks at 13.9 GB and takes 4 per cent longer.

We made 2 earlier folded runs, which show where the time went. Over words of 24 bits, holding one corner arrangement each, the run took 6 h 00. The same run with the depth left as a unary numeral instead of a machine integer took 8 h 13, so the unary depth cost 2 h 13.

The phase one table is 2.9 GB of Rocq source and 4.5 GB once checked, 8 h 48 of processor time. It is generated once and shared with the lower-bound work.

The statement has no hypothesis left.

Theorem `real_row_superflip_fold_runi m :`  
`m \in H -> superflip * m \in ball Sset 20.`

H is the subgroup above, so every position of the superflip’s coset is within 20 moves.

## 6 Conclusion

We prove 2 lower bounds. No position of the cube is solved in 19 face turns, and none in 25 quarter turns. In each case one position is the witness, the superflip for the first and the four-spot with the superflip on it for the second. In each case the proof is a search that comes back empty over a table of estimated distances.

We do not prove that 20 and 26 always suffice. The upper half is a different computation. It is not one search that finds nothing, but 2 billion searches that must each find everything. We did one of the 2 billion, and the section above gives its cost.

The three share a trunk. The cube itself, as permutations of the 48 facelets, with the 18 moves and the group they generate. Balls, the positions within  $d$  moves. The abstract search, whose contract is that a false answer is a proof. The rule that any summary of a position, together with any table that passes one check, gives an estimate that is never too big. And the tables themselves, held as machine integers in arrays rather than as lists of unary numbers.

What each of the three needed of its own:

- **The 20 face turns.** The phase one summary, which is the edge flips and the slice, its table and the certificate that checks the table. The same search seen from 3 angles, and 30 pieces run side by side.
- **The 26 quarter turns.** The cut to 6 prefixes, which is the one piece of the development argued by hand rather than computed, and the parity argument that turns 25 into 24. A second and much larger summary, 29 billion values, with a table of its own and a check of its own.
- **One coset.** A coset held as a map of bits rather than as a tree of positions. That needs 5 new pieces. The rank and the sign of a permutation on machine integers. The link between a position

and its 3 ranks. A pass that steps a whole map one move at a time. The fold by the 16 symmetries. And words given by hand for the members the run does not reach.

The same search written in OCaml is about 4.5 times faster than the one Rocq runs. We ran both at radius 19 on the reference machine. The OCaml program visits 137 607 893 106 positions in 19.2 processor-hours, which is 0.50 microseconds a position. Rocq takes 89.5 processor-hours over the same tree, which is 2.34. A factor of **4.7**.

We do not assume that the two walk the same tree. We divide each of the 30 Rocq pieces by the positions its OCaml counterpart visited. The result is between 1.85 and 2.86 microseconds, over pieces that differ in size by a factor of 2.8. So the run takes a night because the tree holds 138 billion nodes, not because the prover is slow: in OCaml the same tree still costs 19 processor-hours.

The whole development, counted in hand-written Rocq and leaving out the generated tables, is 36 300 lines. Each line of the table counts what that piece adds to the ones above it.

|                                         | <b>files</b> | <b>lines</b>  |
|-----------------------------------------|--------------|---------------|
| the superflip, for the 20 face turns    | 44           | 12 600        |
| the four-spot, for the 26 quarter turns | 19           | 6 100         |
| one coset of the upper bound            | 70           | 17 600        |
| <b>in all</b>                           | <b>133</b>   | <b>36 300</b> |

This development was written with the help of Claude, Anthropic’s coding assistant.

The sources are at [github.com/thery/DoubleCover/code/Rubik](https://github.com/thery/DoubleCover/code/Rubik), with the note, its figures and Reid’s transcribed post beside them.

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